File
Edit
View
Run
Kernel
Tabs
Settings
Help
SALES_19DEC_OVERVIEW.IPYNB
Sales_19Dec_overview.ipyr
Code
Python 3 (ipykernel)

- 1. Preliminery EDA
  - 1.1. Data Overview
  - 1.2. Global metrics
    - 1.2.1. total_uv & total_pv
    - 1.2.2. total_order
    - 1.2.3. GMV
    - 1.2.4. price
    - 1.2.5. event type
  - 1.3. Rawdata Cleaning
- 2. __CUSTOMER ANALYSIS__
  - 2.1. Key Point Indicators KPI
    - 2.1.1. gross conversion rate 转化率
      - 2.1.1.1. funnel model
    - 2.1.2. purchase frequency 用户购买频率
      - 2.1.2.1. requency distribution diagram
    - 2.1.3. repurchase rate 用户复购率
    - 2.1.4. churn cate 用户流失率
  - 2.2. Customer Lifetime Value LTV 用户生命周期价值
    - 2.2.1. Total LTV
    - 2.2.2. Precised LTV by RFM Model
      - 2.2.2.1. histplot of RFM destribution
  - 2.3. User Segmentation 用户分层
    - 2.3.1. User Tags Distribution BarChart
- 3. PLATFROM ANALYSIS
  - 3.1. User Engagement
    - 3.1.1. daily engagement of platfrom
    - 3.1.2. hourly engagement of platfrom
  - 3.2. Conversion Rate in Finer Granularity
    - 3.2.1. total conversion rate
    - 3.2.2. level conversion rate
  - 3.3. Retention Rate
  - 3.4. User Stickiness
- 4. BRAND AND PRODUCT ANALYSIS
  - 4.1. Performance of Brands
    - 4.1.1. highest-heat brand (whichever view/cart/purchase)
    - 4.1.2. outstanding brands
    - 4.1.3. Problematic brands
  - 4.2. Performance of Products
- 5. SUMMARY

[148]:

```python
# -*- coding=utf-8 -*-
# __author = 'bonnieting'__

# basic:
import pandas as pd
import numpy as np
import jieba
import locale
from itertools import product
import warnings
warnings.filterwarnings('ignore')
# time:
from datetime import datetime as dt
import time

# visual:
import matplotlib.pyplot as plt
import matplotlib.ticker as ticker
import plotly.express as px
from plotly import graph_objects as go
import matplotlib.pyplot as plt
import seaborn as sns
# from pyecharts.charts import Funnel
from pyecharts.globals import CurrentConfig, NotebookType
CurrentConfig.NOTEBOOK_TYPE = NotebookType.JUPYTER_LAB
from pyecharts import options as opts
from pyecharts.charts import Bar

# model:
from scipy.stats import norm
from pyecharts import options as opts
from scipy.stats.mstats import winsorize

# dataframe show_way:
# from IPython.display import display
# import os
# import sys
# sys.path.insert(0, '../common')
# from common_htmlTable import HtmlTableMulti
# from IPython.display import display, display_html

locale.setlocale(locale.LC_ALL, 'en_US.UTF-8')
# pd.set_option('display.float_format', '{:.0f}'.format)
np.set_printoptions(precision=4, suppress=True)
pd.set_option('display.float_format',lambda x:'%.2f' %x)
```

[149]:

```python
data=pd.read_csv(r"D:\repo_github\Portfolio1_TMall_sales_EDA\data\2019-Dec.csv")
```

1. Preliminery EDA

1.1. Data Overview

[152]:

```python
data.head()
```

[152]:

	event_time	event_type	product_id	category_id	category_code	brand	price	user_id	user_session
0	2019-12-01 00:00:00 UTC	remove_from_cart	5712790	1487580005268456287	NaN	f.o.x	6.27	576802932	51d85cb0-897f-48d2-918b-ad63965c12dc
1	2019-12-01 00:00:00 UTC	view	5764655	1487580005411062629	NaN	cnd	29.05	412120092	8adff31e-2051-4894-9758-224bfa8aec18
2	2019-12-01 00:00:02 UTC	cart	4958	1487580009471148064	NaN	runail	1.19	494077766	c99a50e8-2fac-4c4d-89ec-41c05f114554
3	2019-12-01 00:00:05 UTC	view	5848413	1487580007675986893	NaN	freedecor	0.79	348405118	722ffea5-73c0-4924-8e8f-371ff8031af4
4	2019-12-01 00:00:07 UTC	view	5824148	1487580005511725929	NaN	NaN	5.56	576005683	28172809-7e4a-45ce-bab0-5efa90117cd5

[153]:

```python
print(data.shape)
print(data.info())
```

```
(3533286, 9)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3533286 entries, 0 to 3533285
Data columns (total 9 columns):
 #   Column         Dtype
---  ------         -----
 0   event_time     object
 1   event_type     object
 2   product_id     int64
 3   category_id    int64
 4   category_code  object
 5   brand          object
 6   price          float64
 7   user_id        int64
 8   user_session   object
dtypes: float64(1), int64(3), object(5)
memory usage: 242.6+ MB
None
```

[154]:

```python
data.isnull().any()
```

[154]:

```
event_time       False
event_type       False
product_id       False
category_id      False
category_code     True
brand             True
price            False
user_id          False
user_session      True
dtype: bool
```

[155]:

```python
data.nunique()
```

[155]:

```
event_time       1654771
event_type             4
product_id         44624
category_id          482
category_code         10
brand                252
price               2122
user_id           370154
user_session      839812
dtype: int64
```

> - ☐ need to see the null in detail
> - ☐ build a scatterplot to see the time distribution of different behaviors.

1.2. Global metrics

1.2.1. total_uv & total_pv

uv: unique visitor ; pv: page view; in total not in each 24hrs

[159]:

```python
event_frequency = data['user_id'].value_counts()
print('event_frequency: \n' + str(event_frequency))
total_uv = len(event_frequency)
print('\ntotal_uv: '+str(total_uv))
total_pv = len(data['user_id'])
print('\ntotal_pv: '+str(total_pv))
```

```
event_frequency:
user_id
527021202    8689
572631412    3883
467810091    3147
548810441    2426
584289031    2356
             ...
561547151       1
586485554       1
586486068       1
363648439       1
595414541       1
Name: count, Length: 370154, dtype: int64

total_uv: 370154

total_pv: 3533286
```

total_order

total closing orders volume 总成交量

[161]:

```python
total_order = len(data[data.event_type=='purchase']['user_session'].value_counts())
print('total_order: '+str(total_order))
#delete invalid column 删除无效列
# data.drop('category_code',axis=1,inplace=True)
```

```
total_order: 28894
```

1.2.3. GMV

gross merchandise volume(ignoring the returns and refunds) 总成交额/gmv（不考虑退货退款）

[163]:

```python
total_gmv = data[data.event_type=='purchase']['price'].sum()
print('total_gmv: '+str(locale.currency(total_gmv,grouping = True))) # f'{total_gmv:.2f}'
```

```
total_gmv: $1,077,624.85
```

[164]:

```python
paid_user = data[data.event_type=='purchase'].nunique()['user_id']
Average_Purchase_Amount = total_gmv/paid_user
print('Average Purchase Amount (mean_amt_per_paid_user): ' + str(locale.currency(Average_Purchase_Amount,grouping = True)))
```

```
Average Purchase Amount (mean_amt_per_paid_user): $42.07
```

price

[166]:

```python
data['price'].describe()
```

[166]:

```
count   3533286.00
mean          8.87
std          19.86
min         -79.37
25%           2.06
50%           4.21
75%           7.14
max         327.78
Name: price, dtype: float64
```

> - ☑ **need to clear the price <0**

event type

categorical statistics on purchase behavior 购买行为分类统计

[169]:

```python
user_event_count = data['event_type'].value_counts()
print(user_event_count)
print(type(user_event_count))
```

```
event_type
view                1728331
cart                 927124
remove_from_cart     664655
purchase             213176
Name: count, dtype: int64
<class 'pandas.core.series.Series'>
```

> - ☑ funnel model
> - ☐ how many views does a customer purchase after that in average?
> - ☐ how many views does a customer purchase the same one product after that in average?

1.3. Rawdata Cleaning

[172]:

```python
data[['user_id','product_id','category_id','user_session']] = data[['user_id','product_id','category_id','user_session']].astype(str)
data.describe()
```

[172]:

	price
count	3533286.00
mean	8.87
std	19.86
min	-79.37
25%	2.06
50%	4.21
75%	7.14
max	327.78

[173]:

```python
data[data['price']<0]['product_id'].value_counts()
```

[173]:

```
product_id
5670257    8
5716859    4
5716857    2
5716855    2
5716861    2
Name: count, dtype: int64
```

> should be the problem of these products. I need to see details by SQL from the product_table in database.
>
> put them into 0 for now.

[175]:

```python
# data['price'] = data['price'].where(data['price']<0, inplace=True)
data['price'] = data['price'].apply(lambda x: max(0, x))
data[data['price']<0]
```

[175]:

event_time	event_type	product_id	category_id	category_code	brand	price	user_id	user_session

[176]:

```python
data[data['price']==0]
```

[176]:

	event_time	event_type	product_id	category_id	category_code	brand	price	user_id	user_session
2586	2019-12-01 01:30:14 UTC	view	5908170	1487580013950664926	NaN	NaN	0.00	579993583	10f5d243-c5a6-4881-8c69-43f64cec9383
2944	2019-12-01 01:49:50 UTC	view	5897797	2195085255034011676	NaN	NaN	0.00	566922379	f3d09cb1-915f-48fc-8749-a83f33e633c1
2971	2019-12-01 01:52:01 UTC	view	5897797	2195085255034011676	NaN	NaN	0.00	566922379	f3d09cb1-915f-48fc-8749-a83f33e633c1
2982	2019-12-01 01:53:00 UTC	view	5897797	2195085255034011676	NaN	NaN	0.00	566922379	43530ca6-3eb1-4f7d-aab6-ebabdae52fe9
3003	2019-12-01 01:55:05 UTC	view	5903915	1487580007675986893	NaN	NaN	0.00	579972058	8c9fb590-38a5-403a-9865-5abd0b09b6c6
...	...	...	...	...	...	...	...	...	...
3529815	2019-12-31 19:47:20 UTC	view	5917178	1487580013950664926	NaN	NaN	0.00	465830873	cd677761-c4b6-4e3f-ae61-45223a807ae7
3531947	2019-12-31 21:59:57 UTC	view	5917178	1487580013950664926	NaN	NaN	0.00	429228723	8f5fc045-535e-4ebf-b19e-a412c2bc99ff
3532329	2019-12-31 22:36:10 UTC	view	5917178	1487580013950664926	NaN	NaN	0.00	595403379	cfc44387-d85d-48d1-ae50-70493f913806
3533004	2019-12-31 23:34:03 UTC	view	5917178	1487580013950664926	NaN	NaN	0.00	337725468	d234964c-d441-4366-af1f-450b8802c87d
3533127	2019-12-31 23:44:33 UTC	view	5917178	1487580013950664926	NaN	NaN	0.00	565040505	9ed2ebf4-a0ca-46fa-a8bf-a89fa13a2c86

7607 rows × 9 columns

> we can see all the price of view type were set as 0 which meant 0 was normal.

[178]:

```python
data[['category_code','brand','user_session']] = data[['category_code','brand','user_session']].fillna('null')
data.isnull().any()
```

[178]:

```
event_time       False
event_type       False
product_id       False
category_id      False
category_code    False
brand            False
price            False
user_id          False
user_session     False
dtype: bool
```

2. **CUSTOMER ANALYSIS**

by user_id

2.1. Key Point Indicators KPI

gross conversion rate 转化率

- *closign ratio 成交率 = unique_paid_user/total_uv*
- *total_paid_rate: What percentage of people who buy, among all the people who browse, add, and buy*

[182]:

```python
view_user = data[data.event_type=='view'].nunique()['user_id']
print('view_user: '+str(view_user))
total_view_rate = (view_user/total_uv)*100
print('total_view_rate: '+str('%.2f'%total_view_rate)+'%')

cart_user = data[data.event_type=='cart'].nunique()['user_id']
print('\ncart_user: '+str(cart_user))
total_cart_rate = (cart_user/total_uv)*100
print('total_cart_rate: '+str('%.2f'%total_cart_rate)+'%')
cart_pct_inview = (cart_user/view_user)*100
print('cart_pct_inview: '+str('%.2f'%cart_pct_inview)+'%')

# pre-defined: paid_user = data[data.event_type=='purchase'].nunique()['user_id']
print('\npaid_user: '+str(paid_user)) # '%.0f'%
total_paid_rate = (paid_user/total_uv)*100
print('total_paid_rate: '+str('%.2f'%total_paid_rate)+'%')
paid_pct_incart = (paid_user/cart_user)*100
print('paid_pct_incart: '+str('%.2f'%paid_pct_incart)+'%')
paid_pct_inview = (paid_user/view_user)*100
print('paid_pct_inview: '+str('%.2f'%paid_pct_inview)+'%')
```

```
view_user: 358212
total_view_rate: 96.77%

cart_user: 83458
total_cart_rate: 22.55%
cart_pct_inview: 23.30%

paid_user: 25613
total_paid_rate: 6.92%
paid_pct_incart: 30.69%
paid_pct_inview: 7.15%
```

> - User behavior flow path:
>
> 1. view->cart->purchase
> 2. view->purchase
> 3. open app->purchase directly
>
> - ☐ group by user_id, product_id, in/ not in one user_session
> - ☐ how long does a customer cost in each link in one user session? -> need more data

funnel model

[252]:

```python
fig = go.Figure(go.Funnel(
    y = ["view", "cart", "purchase"],
    x = [view_user, cart_user, paid_user],
    textposition = "inside",
    textinfo = "value + percent previous",
    opacity = 0.65, marker = {"color": ["lightsalmon", "tan", "teal"],
    "line": {"width": [2, 2, 2], "color": ["wheat", "wheat", "wheat",]}},
    connector = {"line": {"color": "royalblue", "dash": "dot", "width": 3}})
    )
fig.update_layout(width=1000, height=500)
fig.show()
```

> - ☐ need a reference object to decide if the flow percentages are lower than normal ones.(Maybe data from competitive products or the benchmark of a specific field.) -> need more data
> - ☐ Nevertheless, only 23% from view to cart is not enough. I think usually the first thing is to check if there are tremendous diffs between brands which means the product media materials(pics or vedios) matter a lot if yes.
>   - One Step Forward, we can do simple ABTest on materials with Product Department to verify the hypothesis.
>   - Two step forward, we need Marketing Department to do more optimizing the overall quality of materials. For example,
>   - Give good examples of materials
>   - Regularly update with a series of training courses that teach brand stores how to optimize materials well
>   - Emphasize the importance of materials when merchants enter the platform
>   - Or even develop a new backend feature that automatically generates and beautifies materials
> - if it's not materials, it could be:
>   - recommender system needs optimizing: products dont meet their needs.
>   - unfriendly interface: the confusing function button in the converting page
>   - the unreasonable price: like higher than competition
>   - the website experience issues: like loading slowly or a complicated process or unstable connection

purchase frequency 用户购买频率

Distribution of buying times per user 用户购买次数分布

[188]:

```python
user_buy_times = data[data.event_type=='purchase'].groupby('user_id').count()['user_session']
# output Series. should we calculate the frequency in the all "purchase" users or in the all? I think the former making sense.
user_buy_times
```

[188]:

```
user_id
101025416    11
103274988    17
105075440     2
105118203     8
105379623     6
             ..
98079173     14
99011711      4
99293243      7
99499363      8
99666343      1
Name: user_session, Length: 25613, dtype: int64
```

[189]:

```python
user_buy_times.describe()
```

[189]:

```
count   25613.00
mean        8.32
std        10.74
min         1.00
25%         3.00
50%         5.00
75%        10.00
max       399.00
Name: user_session, dtype: float64
```

[190]:

```python
print('max_buying_times: ' + str(user_buy_times.max()))
print('min_buying_times: ' + str(user_buy_times.min()))
Average_Purchase_Frequency = user_buy_times.mean()
print('Average Purchase Frequency (mean_buying_times_per_paid_user): ' + str('%.2f'%Average_Purchase_Frequency))
# buy_times_over20 = user_buy_times.values > 20
# buy_times_over20 = np.count_nonzero(buy_times_over20)
# buy_times_over20 = np.sum(buy_times_over20)
# buy_times_over20
```

```
max_buying_times: 399
min_buying_times: 1
Average Purchase Frequency (mean_buying_times_per_paid_user): 8.32
```

[191]:

```python
buy_times_within_20 = len(user_buy_times[(user_buy_times >= 0) & (user_buy_times <= 20)])
print('Buying_times between [0-20]: ' + str(buy_times_within_20))
buy_times_within_20_ratio = (buy_times_within_20 / paid_user)*100
print("Buying_times between [0-20] ratio: ", str('%.2f'%buy_times_within_20_ratio)+'%')
```

```
Buying_times between [0-20]: 23701
Buying_times between [0-20] ratio:  92.54%
```

requency distribution diagram

to see how many users bought how many times 有多少用户买了XX次

[193]:

```python
bins = np.arange(0, 400, 10)
n, bins, patches = plt.hist(x=user_buy_times, bins=bins, density=False)
plt.xlabel('buy_times')
plt.ylabel('user_number')
plt.title('user_buy_times(excluding the times>130 for users<5)')

bin_centers = 0.5 * (bins[:-1] + bins[1:])
# n = n.astype(float)
freq = n / n.sum()

plt.twinx()
plt.plot(bin_centers, freq, 'r-', linewidth=2.0)
plt.ylabel('frequency')
plt.ylim([0, freq.max() * 1.1])

plt.show()
```

[194]:
```python
print('times data was split into ' + str(len(n)) + ' groups by steplength: 10 .')
freq = freq.tolist()
freq = [f'{f*100:.2f}'+'%' for f in freq]
df1 = pd.DataFrame({'Buying_times From': bins[:-1], 'Buying_times To': bins[1:], 'Frequency': n, 'Normalized Frequency': freq})
df1.head(13)
```

```
times data was split into 39 groups by steplength: 10 .
```

[194]:

	Buying_times From	Buying_times To	Frequency	Normalized Frequency
0	0.00	10.00	18568.00	72.50%
1	10.00	20.00	4929.00	19.24%
2	20.00	30.00	1247.00	4.87%
3	30.00	40.00	426.00	1.66%
4	40.00	50.00	181.00	0.71%
5	50.00	60.00	111.00	0.43%
6	60.00	70.00	60.00	0.23%
7	70.00	80.00	29.00	0.11%
8	80.00	90.00	15.00	0.06%
9	90.00	100.00	15.00	0.06%
10	100.00	110.00	8.00	0.03%
11	110.00	120.00	4.00	0.02%
12	120.00	130.00	5.00	0.02%

> - ☐ can build the VIP system or Key Customers list from the buying_times & buying_amount

repurchase rate 用户复购率

number of users who bought more than once 复购率（购买次数在2次及以上的人数/有过购买记录的人数）

[197]:
```python
twice_buy_user=user_buy_times[user_buy_times>=2].count()
twice_buy_user
rebuy_rate=(twice_buy_user/paid_user)*100
print('rebuy_rate: '+str('%.2f'%rebuy_rate)+'%')
# print("复购率:{:.2f}%".format(rebuy_rate*100))
```

```
rebuy_rate: 88.64%
```

> which means the customer loyalty of this platform is nearly perfect. So the point will be "let customer choose us once.", on the Marketing Deparment to gain trust in the initial expression.
>
> - ☐ how long does a customer cost to rebuy after the first bought in average (to further determine the health) ?
> - ☐ how long between two buying actions of one user in average ?

churn cate 用户流失率

rate of customers who gave up using or buying on our platform 放弃使用、不再活跃的用户比率 reduce churn rate to retain more customers.

- unique users only view / total_uv
- unique users only cart (not to purchase / total_uv
- unique users only buy once / total_uv

[200]:
```python
# we have variable: event_frequency = data['user_id'].value_counts() = data.groupby('user_id').count()['user_session']
once_buy_user = user_buy_times[user_buy_times==1].count()
once_buy_rate = (once_buy_user/total_uv)*100
print('once_buy_churn_rate: '+str('%.2f'%once_buy_rate)+'%')
```

```
once_buy_churn_rate: 0.79%
```

[201]:
```python
event_type_df = data.groupby(['user_id', 'event_type']).count().reset_index()
df3 = event_type_df[event_type_df['event_type']=='view']['user_id']
user_event_type_nums = event_type_df.groupby('user_id').count()['user_session']
user_event_type_nums
only_view_user = len(set(user_event_type_nums[user_event_type_nums==1].index).intersection(set(df3.values)))
only_view_rate = (only_view_user/total_uv)*100
print('only_view_churn_rate: '+str('%.2f'%only_view_rate)+'%')
```

```
only_view_churn_rate: 76.80%
```

[202]:
```python
df5 = event_type_df[event_type_df['event_type']=='cart']['user_id']
cart_no_buy = len(set(df5.values).difference(set(user_buy_times.index)))
cart_no_buy_rate = (cart_no_buy/len(df5))*100
print(str(cart_no_buy) + ' persons in 83458 cart customers didn\'t go to purchase. The only_cart churn rate for the very cart link is: ' + str('%.2f'%cart_no_buy_rate) + '%.')

cart_no_buy_rate_intotal = (cart_no_buy/total_uv)*100
print('only_cart_churn_rate: ' +str('%.2f'%cart_no_buy_rate_intotal) + '%.')
```

```
58168 persons in 83458 cart customers didn't go to purchase. The only_cart churn rate for the very cart link is: 69.70%.
only_cart_churn_rate: 15.71%.
```

[203]:
```python
cart_from_view = len(set(set(df5.values).intersection(set(df3.values))))
cart_from_view_rate = (cart_from_view/len(df5))*100
print(str(cart_from_view) + ' persons in 83458 cart customers came from view. The rate is: ' + str('%.2f'%cart_from_view_rate) + '%.')
```

```
72359 persons in 83458 cart customers came from view. The rate is: 86.70%.
```

[204]:
```python
print('The total churn rate is: ' + str('%.2f'%sum([once_buy_rate, cart_no_buy_rate_intotal, only_view_rate])) + '%.')
```

```
The total churn rate is: 93.30%.
```

> which means the churn rate of **cart_without_buy** and **only_view** is not low though the repurchase rate is highly to 88.64%. So we team should focus on **how to let people try just once**. For reference:
>
> - increasing the conversion rate in user flow to enhance the purchase on the end
> - holding offer campaigns to drawing people to order for the first time
> - guiding customers to make decision: like providing "compare"/"rating" function or lightening the crucial comments
> - collecting feedback and making improvements
>
> cart_without_buy includes customers who view to cart and just cart directly.
>
> **the part in view customers went to purchase is 30.3% from "The only_cart_churn_rate is: 69.70%", less than paid_pct_incart=30.69% in conversion rate chapter by 0.387% because 323 customers purchased directly after view without add to cart, which also means 98.74% of purchase came from cart, a essential function.**

2.2. Customer Lifetime Value LTV 用户生命周期价值

Refined operation of users

2.2.1. Total LTV

*total_LTV = Average Customer Lifespan * Average Purchase Frequency * Average Purchase Amount*

[208]:
```python
event_time_df = data[data.event_type=='purchase'].groupby('user_id')['event_time'].agg(['min', 'max', 'count']).rename(columns={'min':'min_time', 'max':'max_time', 'count':'time_times'}) # last buy time - first buy time = lifespan (not first/last event time
# event_time_df['daysdiff'] = dt.date(event_time_df['max_time']) - dt.date(event_time_df['min_time']) # TypeError: 'Series' object cannot be interpreted as an integer
event_time_df['daysdiff'] = (pd.to_datetime(event_time_df['max_time']) - pd.to_datetime(event_time_df['min_time'])).dt.days + 1
event_time_df = event_time_df.sort_values(by='daysdiff', ascending=False)
event_time_df
```

[208]:

user_id	min_time	max_time	time_times	daysdiff
474001917	2019-12-01 11:56:38 UTC	2019-12-31 14:02:46 UTC	56	31
579991254	2019-12-01 01:42:52 UTC	2019-12-30 05:53:18 UTC	12	30
475372718	2019-12-02 04:02:01 UTC	2019-12-31 17:04:27 UTC	55	30
485997000	2019-12-03 07:10:41 UTC	2019-12-31 17:08:36 UTC	8	29
574105439	2019-12-02 18:01:44 UTC	2019-12-31 00:03:18 UTC	28	29
...	...	...	...	...
525835919	2019-12-24 09:44:55 UTC	2019-12-24 09:44:55 UTC	9	1
525831399	2019-12-05 11:56:53 UTC	2019-12-05 11:56:53 UTC	3	1
525821970	2019-12-28 11:31:11 UTC	2019-12-28 11:31:11 UTC	33	1
525783622	2019-12-21 18:58:08 UTC	2019-12-21 18:58:08 UTC	3	1
99666343	2019-12-17 18:27:15 UTC	2019-12-17 18:27:15 UTC	1	1

25613 rows × 4 columns

[209]:
```python
# effective_event_time_df = event_time_df[event_time_df['daysdiff']>0]
mean_daysdiff = event_time_df['daysdiff'].agg('mean')
print('Average Customer Lifespan: ' + str('%.2f'%mean_daysdiff) + ' days.')
event_time_df['daysdiff'].describe()
# effective_event_time_df['daysdiff'].describe() # same as above
```

```
Average Customer Lifespan: 1.84 days.
```

[209]:
```
count   25613.00
mean        1.84
std         3.44
min         1.00
25%         1.00
50%         1.00
75%         1.00
max        31.00
Name: daysdiff, dtype: float64
```

[210]:
```python
Average_Customer_Lifespan = mean_daysdiff
LTV = Average_Customer_Lifespan * Average_Purchase_Frequency * Average_Purchase_Amount

print('Customer Lifetime Value: $' + str('%.2f'%LTV))
```

```
Customer Lifetime Value: $644.15
```

[211]:
```python
monthly_cvalue = Average_Purchase_Frequency * Average_Purchase_Amount
print('Monthly Customer Value: Per paid customer brought $' + str('%.2f'%monthly_cvalue) + ' in average per month.')
```

```
Monthly Customer Value: Per paid customer brought $350.17 in average per month.
```

[212]:
```python
daysdiff_1 = event_time_df[event_time_df.daysdiff == 1]['daysdiff'].count()
print(daysdiff_1, 'much more than' ,once_buy_user)
```

```
23571 much more than 2910
```

> TIPS
>
> daysdiff == 1 does not mean that he only bought once. run: `data[(data.user_id == 525324399) & (data.event_type == 'purchase')]` .
>
> U'll find he bought 9 times at the same vent time 2019-12-11 20:32:06 UTC. I guess it was a compaign starting poin like Double12 (12, Dec each year). More like robot order manipulation in flash sale.

2.2.2. Precised LTV by RFM Model

segment all customers into 2^3=8 sections by LTV from RFM computing

[215]:
```python
# buy_df = data[data.event_type=='purchase'].groupby('user_id')
event_time_df['max_time2'] = pd.to_datetime(event_time_df['max_time'].str[:-4],utc=True, format='%Y-%m-%d %H:%M:%S').dt.tz_localize(None) # YYYY-MM-DDTHH-MM-SS,, errors='coerce'
analysis_date = event_time_df['max_time2'].max().replace(hour=23, minute=59, second=59) # datetime.today().date() or (event_time_df['max_time2'].max() + pd.DateOffset(days=1)).date()
event_time_df['lastbuy_interval'] =  (analysis_date - event_time_df['max_time2']).dt.days
recency_data = event_time_df['lastbuy_interval'].max() - event_time_df['lastbuy_interval'] # !!I change it into "lifespan" so that the longer lifespan, the higher R-score, that's logical now.

monetary_data = data[data.event_type=='purchase'].groupby('user_id')['price'].sum()

rfm_df = pd.DataFrame({'Recency': recency_data, 'Frequency': user_buy_times, 'Monetary': monetary_data})
rfm_df
```

[215]:

user_id	Recency	Frequency	Monetary
101025416	17	11	62.10
103274988	1	17	60.63
105075440	22	2	11.30
105118203	20	8	39.58
105379623	9	6	20.08
...	...	...	...
98079173	19	14	31.43
99011711	27	4	32.86
99293243	17	7	50.93
99499363	3	8	39.32
99666343	16	1	4.00

25613 rows × 3 columns

> I found the robustness of the 3 dimensions was really high, which meant the model was sensitive to outliers. So I need to exclude outliers first.

[217]:
```python
# exclude extremum values by percentile
recency_pct = rfm_df['Recency'].quantile([0.01, 0.99])
frequency_pct = rfm_df['Frequency'].quantile([0.01, 0.99])
monetary_pct = rfm_df['Monetary'].quantile([0.01, 0.99])

rfm_df = rfm_df[(rfm_df['Recency'] >= recency_pct.iloc[0]) & (rfm_df['Recency'] <= recency_pct.iloc[1])]
rfm_df = rfm_df[(rfm_df['Frequency'] >= frequency_pct.iloc[0]) & (rfm_df['Frequency'] <= frequency_pct.iloc[1])]
rfm_df = rfm_df[(rfm_df['Monetary'] >= monetary_pct.iloc[0]) & (rfm_df['Monetary'] <= monetary_pct.iloc[1])]
```

[218]:
```python
rfm_df['R'], r_bins= pd.qcut(rfm_df['Recency'], q=10, labels=False, retbins=True) # labels=True, retbins=True
rfm_df['F'], f_bins = pd.qcut(rfm_df['Frequency'], q=10, labels=False, retbins=True, duplicates='drop')
rfm_df['M'], m_bins = pd.qcut(rfm_df['Monetary'], q=10, labels=False, retbins=True)

rfm_df['RFM_Score'] = 0.2*rfm_df['R'] + 0.3*rfm_df['F'] + 0.5*rfm_df['M'] # [0.1,0.3,0.6]the weight was decided according to what we cared most.
rfm_df['LTV'] = rfm_df['RFM_Score'] * Average_Purchase_Amount

print("Recency bins:", r_bins)
print("Frequency bins:", f_bins)
print("Monetary bins:", m_bins)
# rfm_df = rfm_df.sort_values(by='LTV', ascending=False)
rfm_df # .reset_index()
```

```
Recency bins: [ 0.  3.  6.  9. 11. 15. 17. 20. 22. 25. 29.]
Frequency bins: [ 1.  2.  3.  4.  5.  7.  9. 12. 17. 50.]
Monetary bins: [  3.67   10.44   13.404  17.3    21.98   28.21   36.012  43.92   54.9
  79.158 248.08 ]
```

[218]:

user_id	Recency	Frequency	Monetary	R	F	M	RFM_Score	LTV
101025416	17	11	62.10	5	6	8	6.80	286.10
103274988	1	17	60.63	0	7	8	6.10	256.65
105075440	22	2	11.30	7	0	1	1.90	79.94
105118203	20	8	39.58	6	5	6	5.70	239.82
105379623	9	6	20.08	2	4	3	3.10	130.43
...	...	...	...	...	...	...	...	...
98079173	19	14	31.43	6	7	5	5.80	244.03
99011711	27	4	32.86	9	2	5	4.90	206.16
99293243	17	7	50.93	5	4	7	5.70	239.82
99499363	3	8	39.32	0	5	6	4.50	189.33
99666343	16	1	4.00	5	0	0	1.00	42.07

24813 rows × 8 columns

[219]:
```python
rfm_df[['Recency', 'Frequency', 'Monetary','LTV']].describe()
```

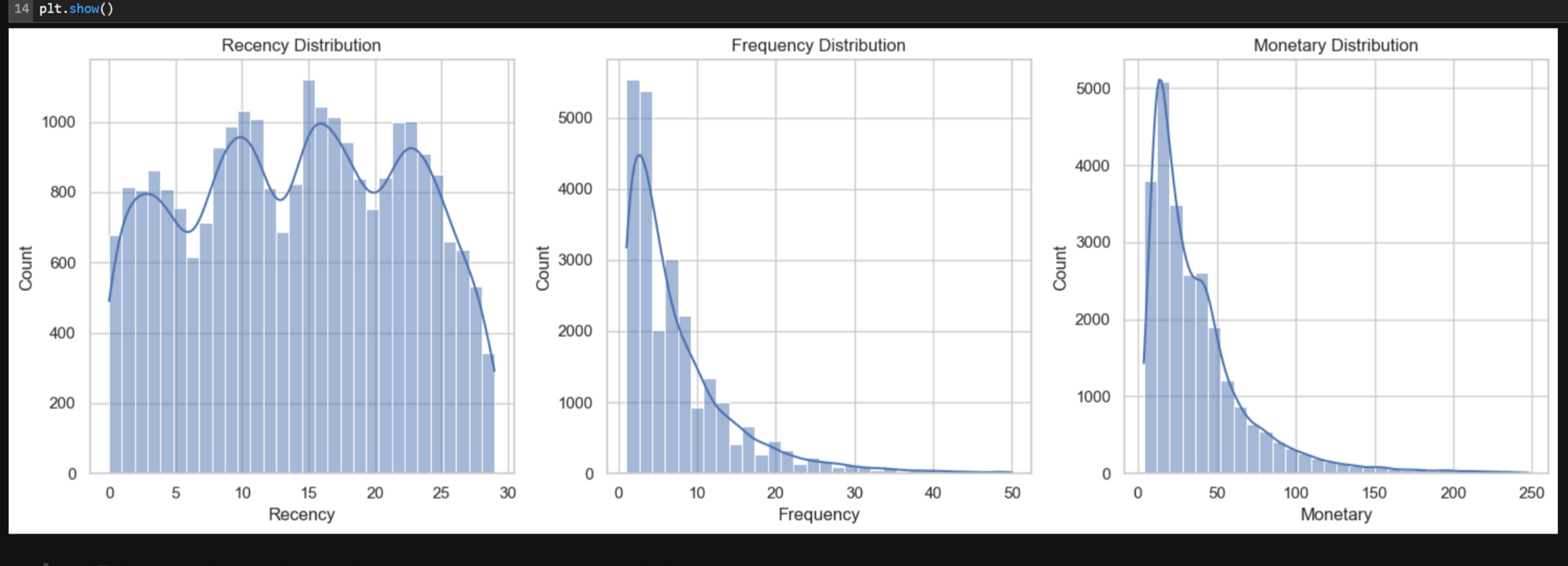
[219]:

	Recency	Frequency	Monetary	LTV
count	24813.00	24813.00	24813.00	24813.00
mean	14.19	7.63	38.37	173.59
std	8.12	7.31	33.66	88.86
min	0.00	1.00	3.67	0.00
25%	8.00	3.00	15.36	100.98
50%	15.00	5.00	28.21	172.50
75%	21.00	10.00	48.41	244.03
max	29.00	50.00	248.08	366.04

> notice that the mean_LTV=366.04 is not a real dollar number but a relative score in total purchase group, a self_defined weight. So it's normal that it != $350.17 from last chapter.

2.2.2.1. histplot of RFM destribution

[222]:
```python
sns.set(style="whitegrid")
fig, axes = plt.subplots(nrows=1, ncols=3, figsize=(15, 5))

sns.histplot(rfm_df['Recency'], bins=30, kde=True, ax=axes[0])
axes[0].set_title('Recency Distribution')

sns.histplot(rfm_df['Frequency'], bins=30, kde=True, ax=axes[1])
axes[1].set_title('Frequency Distribution')

sns.histplot(rfm_df['Monetary'], bins=30, kde=True, ax=axes[2])
axes[2].set_title('Monetary Distribution')

plt.tight_layout()
plt.show()
```



> - ☐ the recency shows that lifespan of customers has obviously periodic. To see insider.

2.3. User Segmentation 用户分层

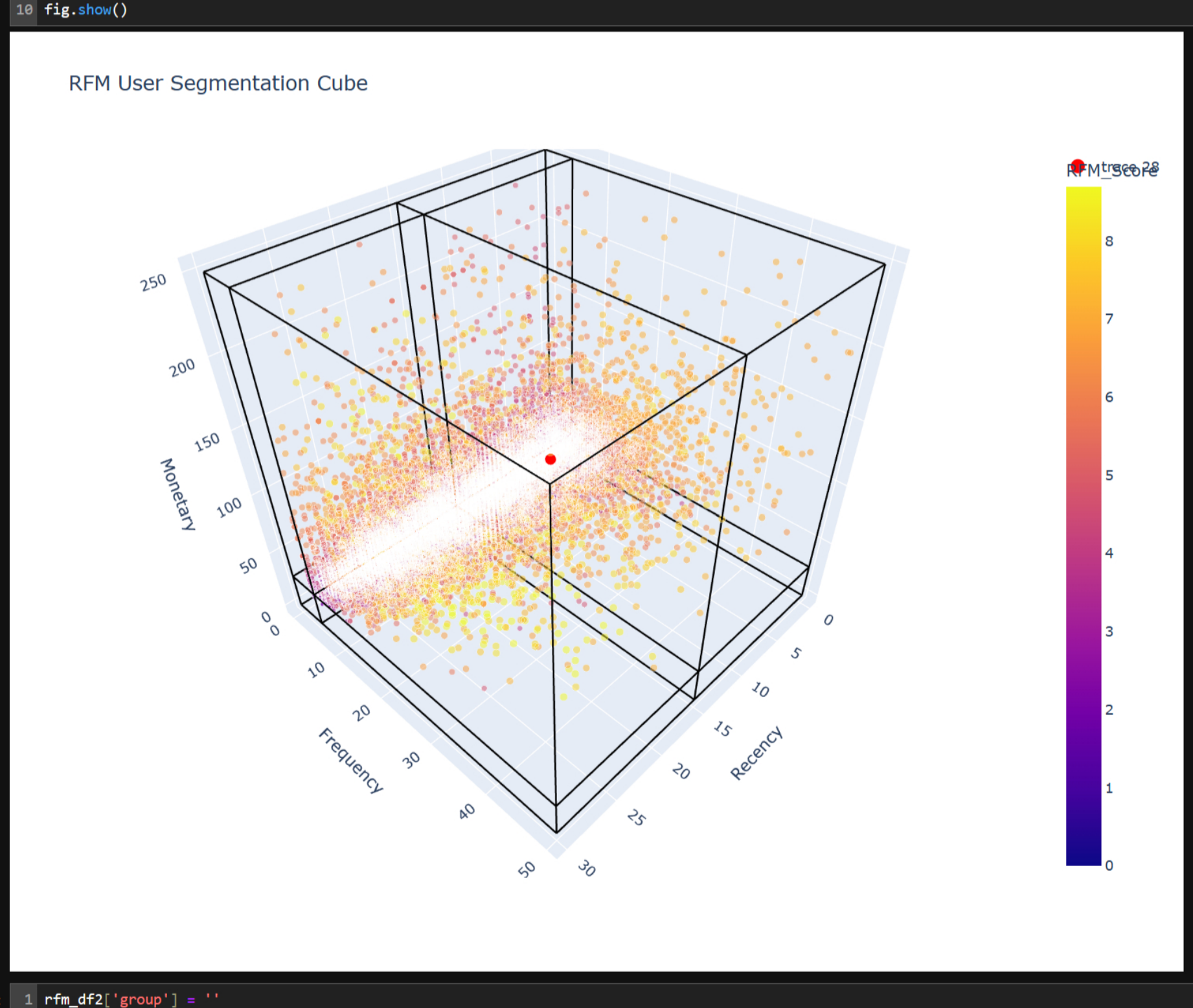
segment all customers into 2^3=8 sections by LTV from RFM computing

[225]:
```python
rfm_df2 = rfm_df.reset_index()
fig = px.scatter_3d(rfm_df2, x='Recency', y='Frequency', z='Monetary', color='RFM_Score',
                    size='RFM_Score', opacity=0.5, size_max=10, hover_name='user_id', #symbol='user_id',
                    title='RFM User Segmentation Cube')

fig.update_layout(scene=dict(xaxis_title='Recency', yaxis_title='Frequency', zaxis_title='Monetary'))
fig.update_layout(width=1000, height=800)

# add straight line to present the 8 subcubes spliting by 50% quantiles above: R-15 F-5 M-28.17
lines_x = [[0,30],[0,30],[0,30],[0,30],[0,30],[0,30],[0,30],[0,30]]
lines_y = [[0,0],[50,50],[0,0],[5,5],[50,50],[0,0],[5,5],[50,50]]
lines_z = [[28.17,28.17],[28.17,28.17],[0,0],[0,0],[0,0],[250,250],[250,250],[250,250]]
for i in range(len(lines_x)):
    fig.add_trace(go.Scatter3d(x=lines_x[i], y=lines_y[i], z=lines_z[i], mode='lines', line=dict(color='black', width=3), name='Line {}'.format(i + 1), showlegend=False))
```

[226]:
```python
lines_x = [[0,0],[30,30],[0,0],[15,15],[30,30],[0,0],[15,15],[30,30]]
lines_y = [[0,50],[0,50],[0,50],[0,50],[0,50],[0,50],[0,50],[0,50]]
lines_z = [[28.17,28.17],[28.17,28.17],[0,0],[0,0],[0,0],[250,250],[250,250],[250,250]]
for i in range(len(lines_x)):
    fig.add_trace(go.Scatter3d(x=lines_x[i], y=lines_y[i], z=lines_z[i], mode='lines', line=dict(color='black', width=3), name='Line {}'.format(i + 1), showlegend=False))
```

[227]:
```python
lines_x = [[15,15],[15,15],[0,0],[0,0],[0,0],[30,30],[30,30],[30,30]]
lines_y = [[0,0],[50,50],[0,0],[5,5],[50,50],[0,0],[5,5],[50,50]]
lines_z = [[0,250],[0,250],[0,250],[0,250],[0,250],[0,250],[0,250],[0,250]]
for i in range(len(lines_x)):
    fig.add_trace(go.Scatter3d(x=lines_x[i], y=lines_y[i], z=lines_z[i], mode='lines', line=dict(color='black', width=3), name='Line {}'.format(i + 1), showlegend=False))
```

[228]:
```python
fig.add_trace(go.Scatter3d(x=[15,15], y=[0,50], z=[28.17,28.17], mode='lines', line=dict(color='black', width=3), showlegend=False)) # legendgroup='divide_R', legendrank=1500 , legendgrouptitle={'text':'Segment'}
fig.add_trace(go.Scatter3d(x=[0, 30], y=[5,5], z=[28.17,28.17], mode='lines', line=dict(color='black', width=3), showlegend=False))
fig.add_trace(go.Scatter3d(x=[15, 15], y=[5,5], z=[0, 250], mode='lines', line=dict(color='black', width=3), showlegend=False))

# add center point
fig.add_trace(go.Scatter3d(x=[15], y=[25], z=[125],
                           mode='markers', marker=dict(color='red', size=5),
                           text=['(15,25,125)'], textposition='middle center'))

fig.show()
```



[229]:
```python
rfm_df2['group'] = ''
for i in range(0,len(rfm_df2)):
    if rfm_df2['R'][i]<5:
        if rfm_df2['F'][i]<5:
            if rfm_df2['M'][i]<5:
                rfm_df2['group'][i]='000'
            else:
                rfm_df2['group'][i]='001'
        else:
            if rfm_df2['M'][i]<5:
                rfm_df2['group'][i]='010'
            else:
                rfm_df2['group'][i]='011'
    else:
        if rfm_df2['F'][i]<5:
            if rfm_df2['M'][i]<5:
                rfm_df2['group'][i]='100'
            else:
                rfm_df2['group'][i]='101'
        else:
            if rfm_df2['M'][i]<5:
                rfm_df2['group'][i]='110'
            else:
                rfm_df2['group'][i]='111'
rfm_df2['group'] = rfm_df2['group'].astype(str)
```

[230]:
```python
rfm_df2
```

[230]:

	user_id	Recency	Frequency	Monetary	R	F	M	RFM_Score	LTV	group
0	101025416	17	11	62.10	5	6	8	6.80	286.10	111
1	103274988	1	17	60.63	0	7	8	6.10	256.65	011
2	105075440	22	2	11.30	7	0	1	1.90	79.94	100
3	105118203	20	8	39.58	6	5	6	5.70	239.82	111
4	105379623	9	6	20.08	2	4	3	3.10	130.43	000
...	...	...	...	...	...	...	...	...	...	...
24808	98079173	19	14	31.43	6	7	5	5.80	244.03	111
24809	99011711	27	4	32.86	9	2	5	4.90	206.16	101
24810	99293243	17	7	50.93	5	4	7	5.70	239.82	101
24811	99499363	3	8	39.32	0	5	6	4.50	189.33	011
24812	99666343	16	1	4.00	5	0	0	1.00	42.07	100

24813 rows × 10 columns

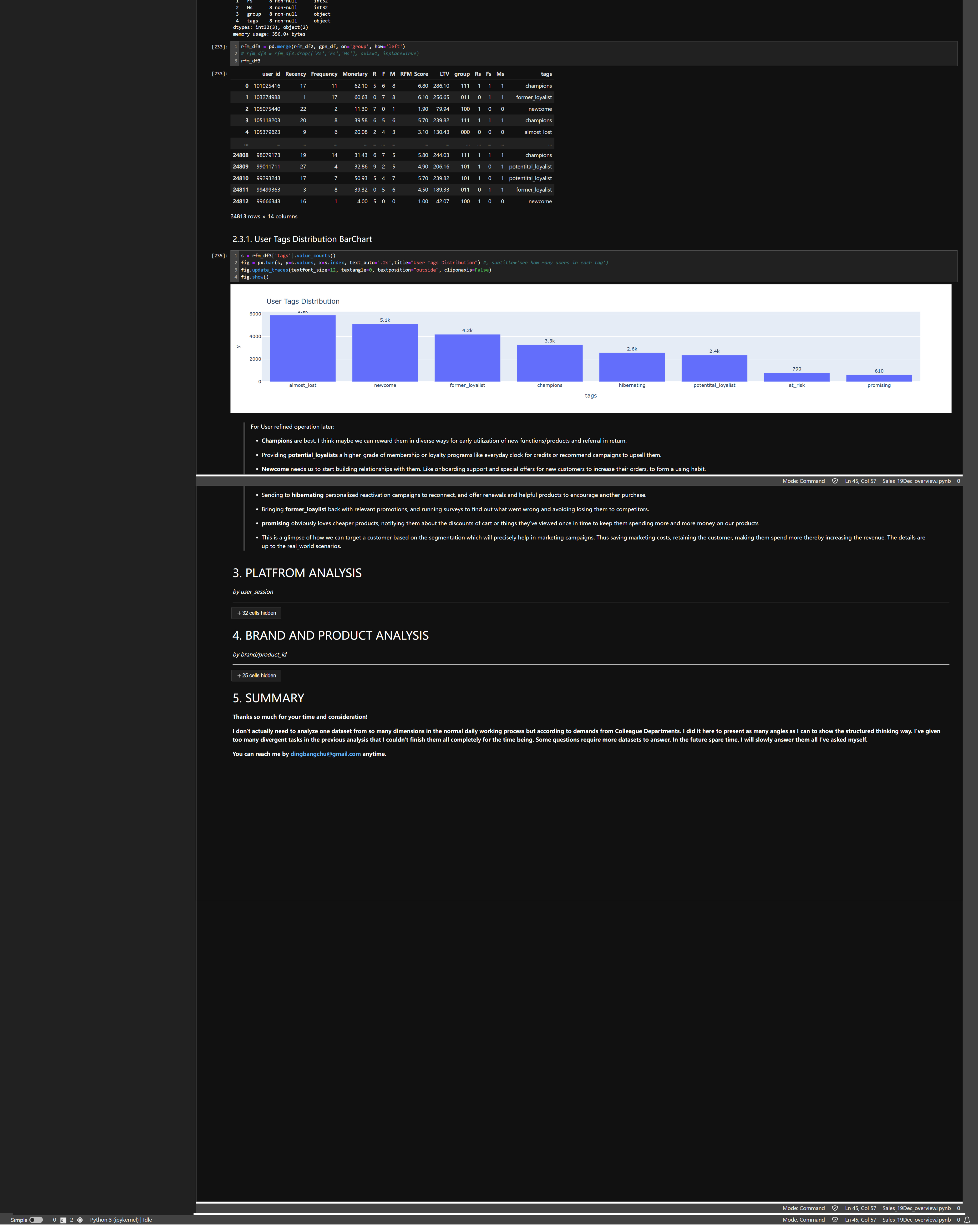
[231]:
```python
a = [1,0]
b = [1,0]
c=[1,0]

comb = list(product(a, b,c))
gpn_df = pd.DataFrame(np.array(comb), columns=['Rs', 'Fs', 'Ms'])
gpn_df['group'] = gpn_df.apply(lambda row: ''.join(map(str, row)), axis=1)
gpn_df['tags'] = ['champions','promising','potentital_loyalist','newcome','former_loyalist','at_risk','hibernating','almost_lost']
gpn_df['group'] = gpn_df['group'].astype(str)
gpn_df
```

[231]:

	Rs	Fs	Ms	group	tags
0	1	1	1	111	champions
1	1	1	0	110	promising
2	1	0	1	101	potentital_loyalist
3	1	0	0	100	newcome
4	0	1	1	011	former_loyalist
5	0	1	0	010	at_risk
6	0	0	1	001	hibernating
7	0	0	0	000	almost_lost

[232]:
```python
gpn_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8 entries, 0 to 7
Data columns (total 5 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Rs      8 non-null      int32
```



```
 1   Fs      8 non-null      int32
 2   Ms      8 non-null      int32
 3   group   8 non-null      object
 4   tags    8 non-null      object
dtypes: int32(3), object(2)
memory usage: 356.0+ bytes
```

[233]:

```python
rfm_df3 = pd.merge(rfm_df2, gpn_df, on='group', how='left')
# rfm_df3 = rfm_df3.drop(['Rs','Fs','Ms'], axis=1, inplace=True)
rfm_df3
```

[233]:

	user_id	Recency	Frequency	Monetary	R	F	M	RFM_Score	LTV	group	Rs	Fs	Ms	tags
0	101025416	17	11	62.10	5	6	8	6.80	286.10	111	1	1	1	champions
1	103274988	1	17	60.63	0	7	8	6.10	256.65	011	0	1	1	former_loyalist
2	105075440	22	2	11.30	7	0	1	1.90	79.94	100	1	0	0	newcome
3	105118203	20	8	39.58	6	5	6	5.70	239.82	111	1	1	1	champions
4	105379623	9	6	20.08	2	4	3	3.10	130.43	000	0	0	0	almost_lost
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
24808	98079173	19	14	31.43	6	7	5	5.80	244.03	111	1	1	1	champions
24809	99011711	27	4	32.86	9	2	5	4.90	206.16	101	1	0	1	potentital_loyalist
24810	99293243	17	7	50.93	5	4	7	5.70	239.82	101	1	0	1	potentital_loyalist
24811	99499363	3	8	39.32	0	5	6	4.50	189.33	011	0	1	1	former_loyalist
24812	99666343	16	1	4.00	5	0	0	1.00	42.07	100	1	0	0	newcome

24813 rows × 14 columns

2.3.1. User Tags Distribution BarChart

[235]:

```python
s = rfm_df3['tags'].value_counts()
fig = px.bar(s, y=s.values, x=s.index, text_auto='.2s',title="User Tags Distribution") #, subtitle='see how many users in each tag')
fig.update_traces(textfont_size=12, textangle=0, textposition="outside", cliponaxis=False)
fig.show()
```

> For User refined operation later:
>
> - **Champions** are best. I think maybe we can reward them in diverse ways for early utilization of new functions/products and referral in return.
> - Providing **potential_loyalists** a higher_grade of membership or loyalty programs like everyday clock for credits or recommend campaigns to upsell them.
> - **Newcome** needs us to start building relationships with them. Like onboarding support and special offers for new customers to increase their orders, to form a using habit.
> - Sending to **hibernating** personalized reactivation campaigns to reconnect, and offer renewals and helpful products to encourage another purchase.
> - Bringing **former_loaylist** back with relevant promotions, and running surveys to find out what went wrong and avoiding losing them to competitors.
> - **promising** obviously loves cheaper products, notifying them about the discounts of cart or things they've viewed once in time to keep them spending more and more money on our products
> - This is a glimpse of how we can target a customer based on the segmentation which will precisely help in marketing campaigns. Thus saving marketing costs, retaining the customer, making them spend more thereby increasing the revenue. The details are up to the real_world scenarios.

3. PLATFROM ANALYSIS

by user_session

---

4. BRAND AND PRODUCT ANALYSIS

by brand/product_id

---

5. SUMMARY

Thanks so much for your time and consideration!

I don't actually need to analyze one dataset from so many dimensions in the normal daily working process but according to demands from Colleague Departments. I did it here to present as many angles as I can to show the structured thinking way. I've given too many divergent tasks in the previous analysis that I couldn't finish them all completely for the time being. Some questions require more datasets to answer. In the future spare time, I will slowly answer them all I've asked myself.

You can reach me by dingbangchu@gmail.com anytime.